Uniform-in-time propagation of chaos for finite nonlinear recombinations with explicit empirical initial error

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Editorial note. The quantitative statement retains the initial empirical Hamming-Wasserstein error. No canonical conditioning rate or unreviewed local central limit theorem is assumed.

Editorial note. The original Section3.4 reuses Q for the full vector field Q(p)-p from the earlier gain-operator convention. Its explicit bilinear formula and displayed vector field1318 determine the meaning.

Editorial note. The variational solution is defined by1323-1330 along q_t=S_t(q). The fixed-base ordinary semigroup display1331-1333 is not selected or used. The generator comparison relies instead on the separately proved identity1786-1821. No replacement flow formula is inserted into the author text.

Editorial note. At1370 the initial term for h_t=bar S_t(q)(p) is determined by h0=p and the following explicit u/v equations1374. The printed q is an initial-value symbol slip; retain the author display and disclose it.

Editorial note. In1856/1860 the coarse error estimates have factor4. Keeping it through the product expansion gives a valid K_s no larger than4k e^(-Ds)(n^5+(k-1)n^4 e^(-Ds))/N^2 for normalized tests. Its integral after multiplyingN/2 is at most2k^2n^5(1-e^(-Dt))/(DN), below the coefficient4 printed in the counted theorem920. The smaller intermediate display1864/1868 is not separately certified.

Editorial note. At1862 the particle pair indices run up to N, as in the generator and1827, not the n sites printed there. Several native displays1511/1780/1801/1832 contain unmatched layout parentheses; any necessary delimiter adaptation must preserve their actual author expressions and be recorded mechanically.

Editorial note. The original source1207-1212 pair-in-V notation means inclusion of both coordinates. Original1172-1179 crosses a left delimiter inside an aligned display; retain mathematics when applying any required layout-only delimiter handling.

Editorial note. Source1674 writes an equality for the expected Wasserstein distance, while the explicitly constructed coupling in1661–1664 only establishes the required upper bound. Equality need not hold because that coupling need not be optimal. The subsequent calculation uses only this upper bound. Preserve

the original formula and read this intermediate step as the already-established inequality; no new proof or changed conclusion is supplied.

Editorial scope. One quantitative empirical-initial-error theorem only. The derivative estimate and W monotonicity are uncounted supporting proofs. The canonical O(N^-1/2) result305-316 and its local-CLT/equivalence-of-ensembles chain are not claimed prepared by this task.

Allowed standard prerequisites (editorial).

- Finite-dimensional polynomial ODE existence, variation equations and finite continuous-time Markov generators. Only general finite-dimensional calculus. The model-specific derivative contraction, second-order bound and generator cancellation are retained local obligations.
- Optimal couplings on finite spaces, convexity of Wasserstein distance, product telescoping and Tonelli. Basic tools; the marked-tree cancellation and uniform-in-time bounds are not supplied as free theorems.
- Sampling with versus without replacement for a symmetric N-particle law. For a bounded k-particle test, replacing sampling of distinct labels by iid labels changes its expectation by at most2*||test||_infinity times the probability of a repeated label, bounded by k(k-1)/(2N).. Justifies the ordinary T1 sampling term; the author uses the looser2k(k-1)/N.

Source reference. Original Theorem 1.5: sources/nonlinear-recombinations-ahl-2024-219/source0.tex, source lines 305–316. The source introduction states the eventual application. This task counts the stronger explicit empirical-error estimate916-924 and proves it without the canonical-initial specialization or local-CLT chain.

Scope of the quantitative propagation estimate. Fix the finite recombination space with $n \geq 2$ sites, a separating measure ν, and N particles. In the displayed estimate take $1 \leq k \leq N$. The probability measures μ_N are symmetric as in the author's definition of chaos. The empirical law, the Hamming–Wasserstein distance and the particle generator are those defined in the original source below. The estimate retains its initial empirical-distance term; it does not assert the separate canonical-initial-data rate.

0.1 The nonlinear equation

Let $\Omega = \prod_{i=1}^{n} X_i$ be the set of n-vectors $\sigma = (\sigma_1, \ldots, \sigma_n)$ where $\sigma_i \in X_i$, and the X_i are given finite sets. We interpret σ as a particle. Thus, a particle is a string of n characters each taken from a finite alphabet. A basic example is obtained by taking $\Omega = \{0,1\}^n$. Without loss of generality we will assume that each X_i has the form $X_i := \{0, 1, 2, \ldots, q_i\}$, for some $q_i \in \mathbb{N}$. Given a subset $A \subset [n]$, $[n] = \{1, \ldots, n\}$, and $\sigma \in \Omega$, σ_A denotes the $A-$component of σ, that is the string $(\sigma_i, i \in A)$. If $(\sigma, \eta) \in \Omega \times \Omega$ is a pair of particles, the recombination at A consists in exchanging the A-component of σ with the A-component of η. This defines the map

$$(\sigma, \eta) \mapsto (\eta_A \sigma_{A^c}, \sigma_A \eta_{A^c}),$$

where $\eta_A \sigma_{A^c}$ denotes the element of Ω with entries η_i for $i \in A$ and σ_i for $i \in A^c = [n] \setminus A$. Let $\mathcal{P}(\Omega)$ denote the set of probability measures on Ω. If the original pair (σ, η) is obtained by sampling independently from $p \in \mathcal{P}(\Omega)$, then the new particle $\eta_A \sigma_{A^c}$ is distributed according to $p_A \otimes p_{A^c}$, the product of the two marginals of p on the A and A^c components respectively. By choosing the set $A \subset [n]$ according to some distribution ν, one obtains the quadratic collision kernel

$$p \mapsto Q(p) = \sum_{A \subset [n]} \nu(A)\, p_A \otimes p_{A^c}.$$

The nonlinear evolution is defined by the dynamical system $\dot{p}_t = Q(p_t) - p_t$, that is

$$\frac{d}{dt}\, p_t = \sum_{A \subset [n]} \nu(A)\, (p_{t,A} \otimes p_{t,A^c} - p_t) \tag{1}$$

with the initial condition $p_0 = p \in \mathcal{P}(\Omega)$. Here $p_t \in \mathcal{P}(\Omega)$ is the distribution of the particle at time t, $p_{t,A}$ denotes its marginal on A and ν is a given probability measure over the subsets of $[n]$. The study of this model starts with the pioneering work of Geiringer [21]; see also [29, 39, 38, 40] for more recent accounts. It is well known that the Cauchy problem associated to (1) has a unique solution for every initial distribution $p \in \mathcal{P}(\Omega)$; see e.g. [40]. Moreover, it is not difficult to see that the evolution preserves the single site marginals, that is $p_{t,i} = p_i$ for all $t \geqslant 0$ and for all $i \in [n]$. We say that the recombination measure ν is *separating* if for any $i, j \in [n]$ there is a positive probability that the random set A with distribution ν separates i and j. Equivalently, if $r(\nu) < 1$ where

$$r(\nu) := \max_{i < j \in [n]} \nu\left(\{i, j\} \subset A \text{ or } \{i, j\} \subset A^c\right) \tag{2}$$

0.2 The particle system

Suppose there are N "particles", described by variables $\eta(j) \in \Omega$, $j = 1, \ldots, N$. That is, each particle is a single string from Ω and $\eta_i(j)$ denotes the content of the jth particle at site $i \in [n]$. We may picture $\eta \in \Omega^N$ as a $N \times n$ matrix such that each row is a particle with n entries, and for each $i \in [n]$, the ith column η_i represents the content of site i for different particles.

Notice that N and n play two very different roles here. The number N of particles will eventually be taken to $+\infty$ to recover the non linear mean field limit, in accordance with the general Kac program. The number n should be thought as a fixed, possibly large quantity describing the size of a single particle space.

The Markov process is given by the following random pair-exchange process. Pairs of particles $\{j, l\}$, $1 \leqslant j < l \leqslant N$ are chosen independently according to a Poisson clock process with rate $1/N$. When the pair $\{j, l\}$ "rings", then a set $A \subset [n]$ is chosen with probability $\nu(A)$ and the recombination

$$(\eta(l), \eta(j)) \mapsto \big(\eta_A(j)\eta_{A^c}(l), \eta_A(l)\eta_{A^c}(j)\big) \tag{3}$$

is performed, that is the A-content is exchanged between particle j and particle l. For all $j, l \in [N], A \subset [n]$, for all $\eta \in \Omega^N$, we write $\eta^{j,l,A}$ for the new

configuration $\eta' \in \Omega$ defined by

$$\eta'(k) = \eta(k),\ \forall\, k \neq j, l; \qquad \eta'(l) = \eta_A(j)\eta_{A^c}(l), \qquad \eta'(j) = \eta_A(l)\eta_{A^c}(j). \quad (4)$$

With this notation the pair configuration in the right hand side of (3) is $(\eta^{j,l,A}(l), \eta^{j,l,A}(j))$. Define also $f^{j,l,A}(\eta) = f(\eta^{j,l,A})$ for all $f : \Omega^N \to \mathbb{R}$. Then, the infinitesimal generator of the continuous time Markov process is given by

$$\mathcal{L}_N f = \frac{1}{N} \sum_{1 \leqslant j < l \leqslant N} \sum_{A \subset [n]} \nu(A) \left(f^{j,l,A} - f \right), \qquad f : \Omega^N \to \mathbb{R}. \quad (5)$$

0.3 Chaos and the propagation of chaos

The chaos property is commonly defined as follows. A measure $\mu_N \in \mathcal{P}(\Omega^N)$ is *symmetric* if it is invariant under any permutation of the N particles. We write $P_k\mu_N \in \mathcal{P}(\Omega^k)$ for the corresponding k-particle marginal.

Definition 0.1 (Kac's chaos or "Boltzmann property"). *A sequence $\mu_N \in \mathcal{P}(\Omega^N)$ of symmetric probabilities on Ω^N is $\mu-$chaotic, for a given $\mu \in \mathcal{P}(\Omega)$, if for any $k \in \mathbb{N}$ one has the weak convergence*

$$P_k\mu_N \longrightarrow \mu^{\otimes k}, \qquad N \to \infty.$$

The main goal in this section is to prove Theorem 1.5. We first introduce some notation. Let $\lambda_\eta \in \mathcal{P}(\Omega)$ be the empirical measure $\lambda_\eta := \frac{1}{N}\sum_{i=1}^N \delta_{\eta(i)}$, where $\eta := (\eta(1), \ldots, \eta(N)) \in \Omega^N$. We consider the Wasserstein distance on $\mathcal{P}(\Omega)$ associated to the Hamming distance on Ω:

$$W(p,q) := \inf_{\Gamma \in \Pi(p,q)} \sum_{\sigma,\, \sigma' \in \Omega} \Gamma(\sigma, \sigma') \sum_{i=1}^n \mathbf{1}_{\sigma_i \neq \sigma'_i}, \quad (6)$$

where $p, q \in \mathcal{P}(\Omega)$, and $\Pi(p,q)$ denotes the set of all couplings Γ of p and q. We remark that the total variation distance $\|p-q\|_{\mathrm{TV}}$ is defined as above with $\sum_{i=1}^n \mathbf{1}_{\sigma_i \neq \sigma'_i}$ replaced by $\mathbf{1}_{\sigma \neq \sigma'}$ and thus one has

$$\|p-q\|_{\mathrm{TV}} \leqslant W(p,q) \leqslant n\, \|p-q\|_{\mathrm{TV}}. \quad (7)$$

In contrast with the total variation distance, the distance W has a convenient monotonicity along the evolution, see Lemma 0.8 and Remark 0.9 below. Recall the definition of the non-separation probability $r(\nu)$ from (2).

0.4 Contractive estimates for the nonlinear semigroup

It is convenient to use the notation $S_t(p) = p_t$ for the solution of (11) with initial datum $p_0 = p$, so that $S_{t+s} = S_tS_s = S_sS_t$, for all $s, t \geqslant 0$. This defines the nonlinear semigroup $\{S_t,\, t \geqslant 0\}$. We show a contraction property for S_t when the two initial distributions p, q have the same marginals, that is $p_i = q_i$ for all $i \in [n]$.

Theorem 0.2. *Assume that $\mu_N \in \mathcal{P}(\Omega^N)$ is $p-$chaotic and the measure ν is separating. Then for all $k \in \mathbb{N}$ and any function of the form $\phi_k := \phi^1 \otimes \cdots \otimes \phi^k$:*

$\Omega^k \mapsto \mathbb{R}$, *where* $\phi^i : \Omega \mapsto \mathbb{R}$, *and such that* $\|\phi_k\|_\infty \leqslant 1$, *the following inequality holds*

$$\begin{aligned}\left|P_k\mu_{N,t}(\phi_k) - p_t^{\otimes k}(\phi_k)\right| \\ \leqslant \frac{2k(k-1)}{N} + \frac{4k^2n^5}{D(\nu)\,N}\left(1 - e^{-D(\nu)\,t}\right) + 2k\,\mu_N\left[W\left(p, \lambda_{\chi_N}\right)\right], \quad (8)\end{aligned}$$

where $\chi_N \in \Omega^N$ *has distribution* μ_N, *and* $D(\nu) := 1 - r(\nu)$.

0.5 Wild sums and McKean trees

Given $f, g : \Omega \to \mathbb{R}$, we adopt the notation

$$(f \circ g)^A := \frac{1}{2}\left(f_A \otimes g_{A^c} + g_A \otimes f_{A^c}\right), \quad (9)$$

$$f \circ g := \sum_{A \subset [n]} \nu(A)(f \circ g)^A, \quad (10)$$

where $f_A(\sigma_A) := \sum_{\sigma_{A^c}} f(\sigma_A\sigma_{A^c})$. Then the nonlinear equation (1) is given by

$$\frac{d}{dt}\,p_t = p_t \circ p_t - p_t. \quad (11)$$

The convolution product defined by $f \circ g$ is commutative and distributive, but not associative. Following Wild's original construction [44] we write the solution of (11) with initial datum $p_0 = p$, as

$$p_t = e^{-t}\sum_{k=1}^{\infty}\left(1 - e^{-t}\right)^{k-1} p^{(k)}, \quad (12)$$

where the $\{p^{(k)}\}_{k\,\geqslant\,1}$ are probability measures on Ω defined inductively by

$$p^{(1)} = p, \qquad p^{(k)} = \frac{1}{k-1}\sum_{j=1}^{k-1} p^{(j)} \circ p^{(k-j)}, \quad k \geqslant 2. \quad (13)$$

The validity of (12) can be easily checked by direct inspection, see e.g. the argument after (49) below for a similar computation.

Moreover, following McKean [31], we may express $p^{(k)}$ as a weighted sum over rooted binary trees γ with k leaves. We now recall the details of this construction, and refer the reader to [7] for further background. Let $\Gamma(k)$ denote the set of rooted binary trees with k leaves and call $\alpha_k(\gamma)$, $\gamma \in \Gamma(k)$, the probability over $\Gamma(k)$ obtained by the following procedure. $\Gamma(1) = \{\gamma_1\}$ is just the empty tree with only the root with $\alpha_1(\gamma_1) = 1$, and $\Gamma(2) = \{\gamma_2\}$ where γ_2 is the unique tree obtained by adding two children to the root, with $\alpha_2(\gamma_2) = 1$. Then, recursively, for $k \geqslant 2$, for any $\gamma_{k-1} \in \Gamma(k-1)$, consider all possible trees $\gamma_{k-1}^i \in \Gamma(k)$, $i = 1, \ldots, k-1$, obtained by adding two children to the ith leaf of γ_{k-1}, and for any $\gamma \in \Gamma(k)$, set

$$\alpha_k(\gamma) = \sum_{\gamma_{k-1}\,\in\,\Gamma(k-1)} \alpha_{k-1}(\gamma_{k-1}) \sum_{i=1}^{k-1} \frac{\mathbf{1}\left(\gamma = \gamma_{k-1}^i\right)}{k-1}\,.$$

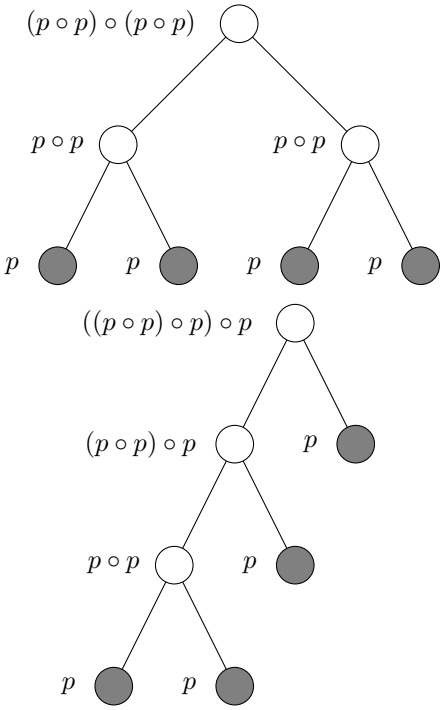


Figure 1: Two possible trees $\gamma, \gamma' \in \Gamma(4)$, and the corresponding distributions $C_\gamma(p) = (p \circ p) \circ (p \circ p)$ and $C_{\gamma'}(p) = ((p \circ p) \circ p) \circ p$.

This defines a probability α_k on $\Gamma(k)$, for any $k \in \mathbb{N}$, and one checks, recursively, that for all $k \geqslant 2$,

$$\alpha_k(\gamma) = \frac{1}{k-1} \alpha_j(\gamma_l) \alpha_{k-j}(\gamma_r), \tag{14}$$

where γ_l and γ_r denote respectively the subtree of γ rooted at the left child of the root and the subtree of γ rooted at the right child of the root, while j denotes the number of leaves in γ_l. Then, by induction over k it follows that for all $k \in \mathbb{N}$,

$$p^{(k)} = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_\gamma(p), \tag{15}$$

where $C_\gamma(p) \in \mathcal{P}(\Omega)$ is described as follows. Each internal node of γ represents a collision and the tree γ describes the collision history. Then $C_\gamma(p)$ represents the distribution obtained at the root after all collisions from γ have been performed, starting with the distribution p at each leaf of γ.

For example, if $k = 3$, there are only two trees $\gamma, \gamma' \in \Gamma(3)$, with $\alpha_3(\gamma) = \alpha_3(\gamma') = 1/2$ and

$$C_\gamma(p) = (p \circ p) \circ p = p \circ (p \circ p) = C_{\gamma'}(p). \tag{16}$$

We refer to Figure 1 for two examples of $C_\gamma(p)$ for $\gamma \in \Gamma(4)$.

Since each collision is of the form $q \circ q'$ for some $q, q' \in \mathcal{P}(\Omega)$ and since by (9) one has

$$p \circ q = \sum_{A \subset [n]} \bar{\nu}(A)\, p_A \otimes q_{A^c}, \tag{17}$$

where $\bar{\nu}(A) = \frac{1}{2}(\nu(A) + \nu(A^c))$, one can write the following expansion for the resulting measure at the root:

$$C_\gamma(p) = \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) C_\gamma^{\vec{A}}(p), \tag{18}$$

where $\mathcal{V}_n$ is the set of subsets of $[n]$, so that $\vec{A} \in \mathcal{V}_n^{k-1}$ represents a pattern $\vec{A} = (A_1, \ldots, A_{k-1})$,

$$\nu(\vec{A}) = \prod_{i=1}^{k-1} \bar{\nu}(A_i) \tag{19}$$

and $C_\gamma^{\vec{A}}(p) \in \mathcal{P}(\Omega)$ is the distribution computed as follows. Each internal node v_i, $i = 1, \ldots, k-1$, in γ (in some fixed order) is associated with the set A_i, and we attach the mark A_i to the edge connecting v_i to its left child and the mark A_i^c to the edge connecting v_i to its right child in γ. In this way we obtain a tree γ with marks on all its edges. Let d_i denote the depth of the ith leaf of γ (e.g. counting from the leftmost leaf), and consider $p_{V_i(\vec{A})}$, the marginal of the measure p on the subset

$$V_i(\vec{A}) := \cap_{j=1}^{d_i} A_j^i, \tag{20}$$

where $A_1^i, \ldots, A_{d_i}^i$ are the marks encountered along the edges of the unique path from the root to the ith leaf. Then $C_\gamma^{\vec{A}}(p)$ is given by

$$C_\gamma^{\vec{A}}(p) = p_{V_1(\vec{A})} \otimes \cdots \otimes p_{V_k(\vec{A})}. \tag{21}$$

Note that some of the $V_i(\vec{A})$ may be empty. However, they form a partition of $[n]$, namely $V_i(\vec{A}) \cap V_j(\vec{A}) = \emptyset$ for $i \neq j$ and $\cup_{i=1}^k V_i(\vec{A}) = [n]$, which can be seen as the result of a fragmentation process

$$[n] \to (A_1, A_1^c) \to (A_1 \cap A_2, A_1 \cap A_2^c, A_1^c) \to \cdots \to \left(V_1(\vec{A}), \cdots, V_k(\vec{A})\right).$$

As an example, in the case (16) we have $\vec{A} = (A_1, A_2)$, and

$$(p \circ p) \circ p = \sum_{A_1,\, A_2 \subset [n]} \bar{\nu}(A_1)\bar{\nu}(A_2) p_{A_1 \cap A_2} \otimes p_{A_1 \cap A_2^c} \otimes p_{A_1^c}. \tag{22}$$

The proof of (15)-(18) can be easily done by induction, by splitting the tree $\gamma \in \Gamma(k)$ into the left and right subtrees γ_l, γ_r and using the relation (14), see the proof of (52) below for a closely related explicit computation.

In conclusion, from (12), (15) and (18) we obtain the following representation of the distribution p_t as a convex combination of distributions $C_\gamma^{\vec{A}}(p)$:

$$p_t = \sum_{k=1}^{\infty} \beta_t(k) \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A})\, C_\gamma^{\vec{A}}(p), \tag{23}$$

where $\beta_t(k) := e^{-t}(1-e^{-t})^{k-1}$ is a probability on $\mathbb{N}$ for each $t \geqslant 0$, $\alpha_k(\gamma)$ is a probability on $\Gamma(k)$ for all k, and $\nu(\vec{A})$ is a probability on $\mathcal{V}_n^{k-1}$ for all k.

Theorem 0.3. *For any probability measure $p, q \in \mathcal{P}(\Omega)$ such that $p_i = q_i$ for all $i \in [n]$, all $t \geqslant 0$,*

$$\|S_t(p) - S_t(q)\|_{\mathrm{TV}} \leqslant \tfrac{1}{2}\, n(n-1)\, \|p-q\|_{\mathrm{TV}}\, e^{-D(\nu)t} \tag{24}$$

where $D(\nu) = 1 - r(\nu)$.

Proof. From (23) we write

$$S_t(p) - S_t(q) = \sum_{k=1}^{\infty} \beta_t(k) s^{(k)}, \tag{25}$$

where

$$s^{(k)} = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \left[C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q) \right]. \tag{26}$$

We introduce some further notation in order to handle more explicitly the difference of probability measures $C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q)$. For any given $\gamma \in \Gamma(k)$, and $\vec{A} \in \mathcal{V}_n^{k-1}$, $\phi : \Omega \mapsto \mathbb{R}$, we may write

$$\begin{aligned} &\langle C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q), \phi \rangle \\ &\quad = \sum_{(x,y) \in \Omega^k \times \Omega^k} \mu(x,y) \sum_{z \in \Omega} \phi(z) \left(\mathbf{1}\left(z = u(y, \vec{A}) \right) - \mathbf{1}\left(z = u(x, \vec{A}) \right) \right) \end{aligned} \tag{27}$$

where $x = (x^i, i = 1, \ldots, k) \in \Omega^k$ stands for the configurations sampled with $q^{\otimes k}$ over the leaves of γ, $y = (y^i, i = 1, \ldots, k) \in \Omega^k$ stands for the configurations sampled with $p^{\otimes k}$ over the leaves of γ, and $\mu(x,y) = \prod_{i=1}^k \mu_1(x^i, y^i)$ denotes a coupling of these two product measures, such that for every $i = 1, \ldots, k$, $\mu_1(x^i, y^i)$ is the optimal coupling of (q, p):

$$\|p-q\|_{\mathrm{TV}} = \sum_{(x,y) \in \Omega^k \times \Omega^k} \mu(x,y) \mathbf{1}\left(x^i \neq y^i\right) = \sum_{x^i, y^i \in \Omega} \mu_1\left(x^i, y^i\right) \mathbf{1}\left(x^i \neq y^i\right). \tag{28}$$

The notation $u(x, \vec{A})$ in (27) is defined as follows. Note that $x^i \in \Omega$ is a vector $x^i = (x^i_1, \ldots, x^i_n)$, for every i, with $x^i_\ell \in \{0, \ldots, q_\ell\}$, and write $x^i_A = (x^i_\ell)_{\ell \in A}$ for the content of x^i on $A \subset [n]$. With this notation, given $x \in \Omega^k$ and $\vec{A} \in \mathcal{V}_n^{k-1}$, $u(x, \vec{A}) \in \Omega$ is defined as the unique configuration such that

$$u(x, \vec{A})_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}$$

for every $i = 1, \ldots, k$. In words, for each $i \in [k]$, the content of the configuration $u(x, \vec{A})$ on the set $V_i(\vec{A}) \subset V$ is taken from the configuration x^i at the ith leaf. The validity of (27) is thus a consequence of (21).

Let $\partial\gamma$ denote the set of leaves of γ. Notice that

$$\mathbf{1}(z = u(x, \vec{A})) = \prod_{i \in \partial\gamma} \mathbf{1}(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})})$$

Let $F = F(\vec{A}, \gamma) \subset \partial\gamma$ denote the set of leaves $i \in \partial\gamma$ such that $|V_i(\vec{A})| > 1$, and write

$$\mathbf{1}(z = u(x, \vec{A})) = w_F(x, z, \vec{A}) w_{F^c}(x, z, \vec{A}),$$

where for any $S \subset \partial\gamma$ we write

$$w_S(x, z, \vec{A}) = \prod_{i \in S} \mathbf{1}(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}). \tag{29}$$

We write $X = (X^i)_{i=1,\dots,k}$ and $Y = (Y^i)_{i=1,\dots,k}$ for the random variables with distribution $q^{\otimes k}$ and $p^{\otimes k}$ respectively. Using the fact that μ is a product over the leaves, and the fact that p, q have the same marginals, for fixed $z \in \Omega$ we have

$$\begin{aligned}\mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] &= \sum_{(x,y)} \mu(x, y) \prod_{i \in F^c} \mathbf{1}\left(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\right) \\ &= \sum_{(x,y)} \mu(x, y) \prod_{i \in F^c} \mathbf{1}\left(z_{V_i(\vec{A})} = y^i_{V_i(\vec{A})}\right) = \mu\left[w_{F^c}(Y, z, \vec{A})\right],\end{aligned} \tag{30}$$

and

$$\begin{aligned}\mu\left[w_{\partial\gamma}\left(Y, z, \vec{A}\right)\right] &= \sum_{(x,y)} \mu(x, y) \mathbf{1}\left(z = u(y, \vec{A})\right) \\ &= \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \mu\left[w_F\left(Y, z, \vec{A}\right)\right].\end{aligned} \tag{31}$$

With this notation we rewrite (27) as

$$\begin{aligned}&\left\langle C^{\vec{A}}_\gamma(p) - C^{\vec{A}}_\gamma(q), \phi \right\rangle \\ &\quad= \sum_{z \in \Omega} \phi(z) \left[\mu[w_{\partial\gamma}\left(Y, z, \vec{A}\right)] - \mu[w_{\partial\gamma}\left(X, z, \vec{A}\right)\right], \\ &\quad= \sum_{z \in \Omega} \phi(z) \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \mu\left[w_F\left(Y, z, \vec{A}\right) - w_F\left(X, z, \vec{A}\right)\right].\end{aligned} \tag{32}$$

Note that $w_F(Y, z, \vec{A})] \neq w_F(X, z, \vec{A})$ implies that there exists $i \in F$ such that $X^i \neq Y^i$. Therefore, from (28) we see that

$$\begin{aligned}&\sum_{z \in \Omega} \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \left|\mu\left[w_F\left(Y, z, \vec{A}\right) - w_F\left(X, z, \vec{A}\right)\right]\right| \\ &\leqslant \sum_{i \in F} \sum_{z \in \Omega} \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \mu\left[\left|w_F\left(Y, z, \vec{A}\right) - w_F\left(X, z, \vec{A}\right)\right| \mathbf{1}\left(X^i \neq Y^i\right)\right] \\ &\leqslant 2 \sum_{i \in F} \mu\left[X^i \neq Y^i\right] = 2|F| \, \|p - q\|_{\mathrm{TV}}.\end{aligned} \tag{33}$$

Therefore, we have shown that (26) satisfies

$$\left|\left\langle s^{(k)}, \phi \right\rangle\right| \leqslant 2 \|\phi\|_\infty \|p - q\|_{\mathrm{TV}} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \, |F|. \tag{34}$$

To estimate (34), let us call $\mathcal{A}_j = \mathcal{A}_j(\gamma)$ the set of $\vec{A} \in \mathcal{V}_n^{k-1}$ such that $|V_j(\vec{A})| \geqslant 2$, and let $\mathcal{A} = \cup_{j=1}^k \mathcal{A}_j$. If $\vec{A} \notin \mathcal{A}$ then all sets $V_j(\vec{A})$ are either empty or a single site and therefore $|F| = 0$. Moreover, $|F| = \sum_{j=1}^k \mathbf{1}(\mathcal{A}_j)$. Therefore we write

$$\sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A})\, |F| = \sum_{j=1}^k \nu(\mathcal{A}_j). \tag{35}$$

To estimate the probability $\nu(\mathcal{A}_j)$, observe that the event $\mathcal{A}_j$ implies that there exists $i_1 < i_2 \in [n]$ such that $\{i_1, i_2\} \in V_j(\vec{A})$. Moreover, recalling (20),

$$\nu\left(\{i_1, i_2\} \in V_j(\vec{A})\right) = \nu\left(\{i_1, i_2\} \in \cap_{l=1}^{d_j} A_l^j\right) = \bar{\nu}\left(\{i_1, i_2\} \in A\right)^{d_j} \leqslant \left(\frac{r(\nu)}{2}\right)^{d_j},$$

where we use the fact that for any $i_1 < i_2$, the probability under $\bar{\nu}$ that $\{i_1, i_2\} \in A$ is bounded by $r(\nu)/2$. Indeed, conditionally on not being separated the probability that both i_1, i_2 belong to A under $\bar{\nu}$ is $1/2$ by symmetry. A union bound then shows that

$$\nu(\mathcal{A}_j) \leqslant \tfrac{1}{2}\, n(n-1) \left(\frac{r(\nu)}{2}\right)^{d_j}.$$

In conclusion, if we define

$$\omega(\gamma) := \sum_{j=1}^k \left(\frac{r(\nu)}{2}\right)^{d_j}, \tag{36}$$

we obtain, for all $k \in \mathbb{N}$,

$$\left|\left\langle s^{(k)}, \phi \right\rangle\right| \leqslant n(n-1)\, \|\phi\|_\infty\, \|p - q\|_{\mathrm{TV}} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\, \omega(\gamma). \tag{37}$$

From [7, Lemma 1.4], one has, for all $\varepsilon > 0$,

$$\sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\omega(\gamma) \leqslant B_\varepsilon e^{-a_\varepsilon \log k}. \tag{38}$$

with $a_\varepsilon = (1 - r(\nu))/(1 + \varepsilon)$ and $B_\varepsilon = B_\varepsilon(r(\nu), \varepsilon)$. One could use this estimate to obtain an almost optimal decay rate a_ε. However, here we obtain the optimal exponential decay rate by computing the expectation of $\omega(\gamma)$.

Lemma 0.4. *For any $t \geqslant 0$,*

$$\sum_{k=1}^\infty \beta_t(k) \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\, \omega(\gamma) = e^{-(1-r(\nu))t}. \tag{39}$$

Assuming the validity of (39), and using $\|\mu - \mu'\|_{\mathrm{TV}} = \frac{1}{2} \max_{f:\, \|f\|_\infty \leqslant 1} |\mu(f) - \mu'(f)|$, from (25) and (37) we obtain

$$\|S_t(p) - S_t(q)\|_{\mathrm{TV}} \leqslant \tfrac{1}{2}\, n(n-1)\, \|p - q\|_{\mathrm{TV}}\, e^{-(1-r(\nu))t}, \tag{40}$$

which concludes the proof of Theorem 0.3.

It remains to prove (39). For any $\gamma \in \cup_{k=1}^{\infty}\Gamma(k)$, define $\mathbb{P}_t(\gamma) = \beta_t(k)\alpha_k(\gamma)$, where k is such that $\gamma \in \Gamma(k)$. Notice that for any $\gamma \in \cup_{k=1}^{\infty}\Gamma(k)$,

$$\mathbb{P}_t(\gamma) = \mathbf{1}_{\gamma=\emptyset}\, e^{-t} + \mathbf{1}_{\gamma\neq\emptyset} \int_0^t e^{-s}\mathbb{P}_{t-s}(\gamma_l)\mathbb{P}_{t-s}(\gamma_r)ds, \tag{41}$$

where $\emptyset$ denotes the empty tree with one leaf (given by the root), while for $\gamma \neq \emptyset$ we write γ_l, γ_r for the left and right subtrees. Indeed, for $\gamma = \emptyset$ (41) is immediate, while for $\gamma \neq \emptyset$, supposing $\gamma \in \Gamma(k)$, and recalling (14) one has

$$\begin{aligned}\mathbb{P}_t(\gamma) &= e^{-t}\left(1-e^{-t}\right)^{k-1}\alpha_k(\gamma) = e^{-t}\int_0^t e^{-(t-s)}\left(1-e^{-(t-s)}\right)^{k-2}(k-1)\alpha_k(\gamma)ds \\ &= \int_0^t e^{-s}e^{-2(t-s)}\left(1-e^{-(t-s)}\right)^{k-2}\alpha_j(\gamma_l)\alpha_{k-j}(\gamma_r)ds = \int_0^t e^{-s}\mathbb{P}_{t-s}(\gamma_l)\mathbb{P}_{t-s}(\gamma_r)ds.\end{aligned}$$

To prove Lemma 0.4, note that the left hand side of (39) is given by $\sum_\gamma \mathbb{P}_t(\gamma)\omega(\gamma)$. Using (41) and $\omega(\emptyset) = 1$, we see that

$$\begin{aligned}\zeta(t) :&= \sum_\gamma \mathbb{P}_t(\gamma)\omega(\gamma) = e^{-t} + \frac{r(\nu)}{2}\sum_{\gamma\neq\emptyset}\int_0^t e^{-s}\mathbb{P}_{t-s}(\gamma_l)\mathbb{P}_{t-s}(\gamma_r)(\omega(\gamma_l)+\omega(\gamma_r))ds \\ &= e^{-t} + r(\nu)\sum_\gamma \int_0^t e^{-s}\mathbb{P}_{t-s}(\gamma)\omega(\gamma)ds = e^{-t} + r(\nu)\int_0^t e^{-s}\zeta(t-s)ds.\end{aligned}$$

Differentiating, and integrating by parts the resulting expression one finds $\dot\zeta(t) = -(1-r(\nu))\zeta(t)$. Since $\zeta(0) = 1$ we obtain $\zeta(t) = e^{-(1-r(\nu))t}$, $t \geqslant 0$. □

0.6 Contraction for the linearized equation

Consider the symmetric bilinear form $\widehat{Q}(f,g)$ defined by

$$\begin{aligned}&\widehat{Q}(f,g)(\eta) \\ &= \frac{1}{2}\sum_{A,\sigma}\nu(A)\big(f(\sigma_A\eta_{A^c})g(\eta_A\sigma_{A^c})+g(\sigma_A\eta_{A^c})f(\eta_A\sigma_{A^c})-f(\sigma)g(\eta)-g(\sigma)f(\eta)\big),\end{aligned} \tag{42}$$

where $\eta \in \Omega$, $f, g : \Omega \mapsto \mathbb{R}$, and the sum extends to all $A \subset [n]$ and $\sigma \in \Omega$. If $p \in \mathcal{P}(\Omega)$, then $\widehat{Q}(p,p) = Q(p)$, where

$$Q(p) = \sum_A \nu(A)(p_A \otimes p_{A^c} - p)$$

is the generator associated to (1), that is the non linear semigroup $\{S_t, t \geqslant 0\}$ satisfies $S_0(p) = p$ and $\partial_t S_t(p) = Q(S_t(p))$, $t \geqslant 0$. Next, consider the differential equation

$$\partial_t h_t = 2\widehat{Q}(q_t, h_t), \qquad h_0 = h, \tag{43}$$

where $q_t = S_t(q)$ for some fixed $q \in \mathcal{P}(\Omega)$. Note that it is linear in h. We write its unique solution as

$$h_t = \bar{S}_t(q)(h),$$

that is $\bar{S}_t(q)(h)$ verifies $\partial_t \bar{S}_t(q)(h) = 2\widehat{Q}(q_t, \bar{S}_t(q)(h))$, $\bar{S}_0(q)(h) = h$ and

Our main result concerning the linearized equation reads as follows.

Theorem 0.5. *For all $p, q \in \mathcal{P}(\Omega)$ such that $p_i = q_i$ for all $i \in [n]$, and $\phi : \Omega \mapsto \mathbb{R}$, for all $t > 0$,*

$$\left\langle \bar{S}_t(q)(p-q), \phi \right\rangle \leqslant n(n-1)\|\phi\|_\infty e^{-D(\nu)\, t}\, \|p-q\|_{\mathrm{TV}}, \tag{44}$$

$$\begin{aligned}\left\langle S_t(p) - S_t(q) - \bar{S}_t(q)(p-q), \phi \right\rangle \\ \leqslant \frac{1}{16}\, n^3(n-1)(n-2)\|\phi\|_\infty e^{-D(\nu)\, t}\, \|p-q\|_{\mathrm{TV}}^2,\end{aligned} \tag{45}$$

where $D(\nu) = 1 - r(\nu)$.

The proof of Theorem 0.5 requires several steps. We start by giving an explicit representation of the solution $\bar{S}_t(q)(p-q)$ to (43) when $h = p - q$. To this end let us define, for any $k \geqslant 2$, $\gamma \in \Gamma(k)$, the distribution $C_{\gamma,p}(q)$ as the measure defined in (18) with the only difference that we take the distribution p instead of q at the rightmost leaf of γ, all other leafs remaining with the distribution q. Formally,

$$C_{\gamma,p}(q) = \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A})\, C^{\vec{A}}_{\gamma,p}(q), \qquad C^{\vec{A}}_{\gamma,p}(q) = q_{V_1(\vec{A})} \otimes \cdots \otimes q_{V_{k-1}(\vec{A})} \otimes p_{V_k(\vec{A})}. \tag{46}$$

In what follows we denote by $d_r(\gamma)$ the depth of the rightmost leaf in γ.

Lemma 0.6. *For all $p, q \in \mathcal{P}(\Omega)$, $t \geqslant 0$,*

$$\bar{S}_t(q)(p-q) = \sum_{k=1}^{\infty} \beta_t(k) \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\, (C_{\gamma,p}(q) - C_\gamma(q))\, 2^{d_r(\gamma)}. \tag{47}$$

Proof. Fix $p, q \in \mathcal{P}(\Omega)$. If $h_t = \bar{S}_t(q)(p)$, then

$$\partial_t \sum_\sigma h_t(\sigma) = 2 \sum_\sigma \widehat{Q}(q_t, h_t)(\sigma) = 0.$$

Thus $\sum_\sigma h_t(\sigma) = 1$, and $2\widehat{Q}(q_t, h_t) = 2q_t \circ h_t - q_t - h_t$. Therefore h_t satisfies

$$h_t = qe^{-t} + \int_0^t e^{-(t-s)}\, (2q_s \circ h_s - q_s)\, ds. \tag{48}$$

By linearity, $\bar{S}_t(q)(p-q) = \bar{S}_t(q)(p) - \bar{S}_t(q)(q)$ satisfies $\bar{S}_t(q)(p-q) = u_t - v_t$, where

$$u_t = pe^{-t} + \int_0^t e^{-(t-s)} 2q_s \circ u_s ds, \qquad v_t = qe^{-t} + \int_0^t e^{-(t-s)} 2q_s \circ v_s ds. \tag{49}$$

Wild's construction then shows that

$$u_t = e^{-t} \sum_{k=1}^{\infty} \left(1 - e^{-t}\right)^{k-1} u^{(k)}, \qquad v_t = e^{-t} \sum_{k=1}^{\infty} \left(1 - e^{-t}\right)^{k-1} v^{(k)}, \tag{50}$$

where $u^{(1)} = p$, $v^{(1)} = q$, and for $k \geqslant 2$,

$$u^{(k)} = \frac{1}{k-1} \sum_{j=1}^{k-1} 2q^{(j)} \circ u^{(k-j)}\,, \qquad v^{(k)} = \frac{1}{k-1} \sum_{j=1}^{k-1} 2q^{(j)} \circ v^{(k-j)}.$$

respectively, and $q^{(j)}$ is defined by (13) with p replaced by q. To check that this representation of the solution holds, let u_t be defined by (50). Then

$$\begin{aligned}
\int_0^t e^{-(t-s)} 2q_s \circ u_s ds &= \sum_{\ell,k=1}^{\infty} \int_0^t \beta_s(k)\beta_s(\ell) e^{-(t-s)} 2q^{(\ell)} \circ u^{(k)} ds \\
&= \sum_{\ell < k} \int_0^t e^{-2s} \left(1 - e^{-s}\right)^{k-2} e^{-(t-s)} 2q^{(\ell)} \circ u^{(k-\ell)} ds \\
&= \sum_{k \,\geqslant\, 2} \int_0^t (k-1) e^{-s} \left(1 - e^{-s}\right)^{k-2} e^{-t} u^{(k)} ds \\
&= \sum_{k \,\geqslant\, 2} \left(1 - e^{-t}\right)^{k-1} e^{-t} u^{(k)} = u_t - e^{-t} u^{(1)} = u_t - e^{-t} p.
\end{aligned} \tag{51}$$

Thus u_t solves (49). The same applies to v_t. This proves (50).

Next, we prove that

$$u^{(k)} = \sum_{\gamma \,\in\, \Gamma(k)} \alpha_k(\gamma) C_{\gamma,p}(q) 2^{d_r(\gamma)}\,, \qquad v^{(k)} = \sum_{\gamma \,\in\, \Gamma(k)} \alpha_k(\gamma) C_{\gamma}(q) 2^{d_r(\gamma)} \tag{52}$$

for all $k \geqslant 2$. Let us prove this by using induction over k. If $k = 2$ then $u^{(2)} = 2q \circ p$ and thus the claimed identity holds. If the formula holds for $j \leqslant k-1$, then using (18) and (14),

$$\begin{aligned}
u^{(k)} &= \frac{1}{k-1} \sum_{j=1}^{k-1} 2q^{(j)} \circ u^{(k-j)} \\
&= \frac{1}{k-1} \sum_{j=1}^{k-1} \sum_{\substack{\gamma_l \,\in\, \Gamma(j) \\ \gamma_r \,\in\, \Gamma(k-j)}} \alpha_j(\gamma_l)\alpha_{k-j}(\gamma_r) C_{\gamma_l}(q) \circ C_{\gamma_r,p}(q) 2^{d_r(\gamma_r)+1} \\
&= \sum_{\gamma \,\in\, \Gamma(k)} \alpha_k(\gamma) C_{\gamma,p}(q) 2^{d_r(\gamma)}.
\end{aligned}$$

This proves the identity (52) for u, and the same argument proves the one for v. The identities (50) and (52) imply (47). □

Next, for any $k \geqslant 2$, $\gamma \in \Gamma(k)$, $j \in [k]$, we define

$$C_{\gamma,p,j}(q) = \sum_{\vec{A} \,\in\, \mathcal{V}_n^{k-1}} \nu(\vec{A}) C^{\vec{A}}_{\gamma,p,j}(q)\,, \qquad C^{\vec{A}}_{\gamma,p,j}(q) := p_{V_j(\vec{A})} \otimes_{\ell \neq j} q_{V_\ell(\vec{A})}, \tag{53}$$

which denotes the distribution obtained from the tree $\gamma \in \Gamma(k)$ when all leaves are given the distribution q except for the jth leaf which takes the distribution p. When $j = k$, that is j is the rightmost leaf, then we have $C_{\gamma,p,k}(q) = C_{\gamma,p}(q)$.

Lemma 0.7. *For any $k \geqslant 2$, for all p, q:*

$$\sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) 2^{d_r} C_{\gamma,p}(q) = \sum_{j=1}^{k} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_{\gamma,p,j}(q)\,. \tag{54}$$

Proof. By symmetry, it holds for $k = 2$, since in this case (54) says $2q \circ p = q \circ p + p \circ q$. Assume that (54) holds for all $j \leqslant k-1$. Then

$$\begin{aligned}
&\sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) 2^{d_r(\gamma)} C_{\gamma,p}(q) \\
&= \frac{2}{k-1} \sum_{j=1}^{k-1} \sum_{\gamma \in \Gamma(j)} \alpha_j(\gamma) C_\gamma(q) \circ \left(\sum_{\gamma' \in \Gamma(k-j)} \alpha_{k-j}(\gamma') 2^{d_r(\gamma')} C_{\gamma',p}(q) \right) \\
&= \frac{2}{k-1} \sum_{j=1}^{k-1} \sum_{\gamma \in \Gamma(j)} \alpha_j(\gamma) C_\gamma(q) \circ \left(\sum_{i=1}^{k-j} \sum_{\gamma' \in \Gamma(k-j)} \alpha_{k-j}(\gamma') C_{\gamma',p,i}(q) \right) \qquad (55) \\
&= 2 \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=j(\gamma)+1}^{k} C_{\gamma,p,i}(q),
\end{aligned}$$

where $j(\gamma)$ is the number of leaves in the left subtree of γ, and we have used (14). By symmetry,

$$\sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=j(\gamma)+1}^{k} C_{\gamma,p,i}(q) = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=1}^{j(\gamma)} C_{\gamma,p,i}(q)\,. \tag{56}$$

Therefore,

$$\begin{aligned}
&2 \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=j(\gamma)+1}^{k} C_{\gamma,p,i}(q) \\
&\qquad = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=1}^{j(\gamma)} C_{\gamma,p,i}(q) + \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=j(\gamma)+1}^{k} C_{\gamma,p,i}(q) \\
&\qquad = \sum_{i=1}^{k} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_{\gamma,p,i}(q).
\end{aligned} \tag{57}$$

The desired identity follows from (55) and (57). □

We can now turn to the proof Theorem of 0.5.

Proof of (44)**.** From Lemma 0.6 and Lemma 0.7, we write

$$\bar{S}_t(q)(p-q) = \sum_{k=1}^{\infty} \beta_t(k)\, \bar{s}^{(k)}, \tag{58}$$

where

$$\bar{s}^{(k)} = \sum_{j=1}^{k} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \left[C^{\vec{A}}_{\gamma,p,j}(q) - C^{\vec{A}}_{\gamma}(q) \right]. \tag{59}$$

Let $\mathcal{A}_j$ denote the set of $\vec{A} \in \mathcal{V}_n^{k-1}$ such that $|V_j(\vec{A})| \geqslant 2$. Since p, q have the same marginals, arguing as in the proof of Theorem 0.3, we obtain, or all $\phi : \Omega \mapsto \mathbb{R}$,

$$\left|\left\langle C^{\vec{A}}_{\gamma,p,j}(q) - C^{\vec{A}}_{\gamma}(q), \phi \right\rangle\right| \leqslant 2\|\phi\|_\infty \|p-q\|_{\mathrm{TV}}\, \mathbf{1}(\mathcal{A}_j)\,. \tag{60}$$

Therefore, as in (37) we obtain

$$\left|\left\langle \bar{s}^{(k)}, \phi \right\rangle\right| \leqslant n(n-1)\,\|\phi\|_\infty\, \|p-q\|_{\mathrm{TV}} \sum_{\gamma\,\in\,\Gamma(k)} \alpha_k(\gamma)\omega(\gamma). \tag{61}$$

From (58) and Lemma 0.4 we conclude the proof of (44). □

Proof of (45)**.** The proof of (45) requires a bit more work. Let us define

$$r_t := S_t(p) - S_t(q) - \bar{S}_t(q)(p-q).$$

From Lemma 0.6 and Lemma 0.7 we see that

$$r_t = \sum_{k=1}^{\infty} \beta_t(k) r^{(k)}, \tag{62}$$

where

$$r^{(k)} = \sum_{\gamma\,\in\,\Gamma(k)} \alpha_k(\gamma) \left[C_\gamma(p) - C_\gamma(q) - \sum_{i=1}^{k} \left(C_{\gamma,p,i}(q) - C_\gamma(q)\right)\right], \tag{63}$$

We are going to prove that for any $\phi : \Omega \mapsto \mathbb{R}$,

$$\left\langle r^{(k)}, \phi \right\rangle \leqslant \tfrac{1}{16}\, n^3(n-1)(n-2)\|\phi\|_\infty \|p-q\|^2_{\mathrm{TV}} \sum_{\gamma\,\in\,\Gamma(k)} \alpha_k(\gamma)\omega(\gamma). \tag{64}$$

By the argument in (40) and Lemma 0.4 this is sufficient to end the proof.

With the notation from the proof of Theorem 0.3 we write

$$\begin{aligned}\left\langle r^{(k)}, \phi \right\rangle = \sum_{\gamma\,\in\,\Gamma(k)} \alpha_k(\gamma) \sum_{(x,y)\,\in\,\Omega^k\times\Omega^k} \sum_{\vec{A}\,\in\,\mathcal{V}_n^{k-1}} \mu(x,y)\nu(\vec{A}) \\ \times \sum_{z\,\in\,\Omega} \phi(z) \left(\mathbf{1}\left(z = u(y,\vec{A})\right) - \mathbf{1}\left(z = u(x,\vec{A})\right) \right. \\ \left. - \sum_{i=1}^{k} \left(\mathbf{1}\left(z = u(x,y,\vec{A},i\right) - \mathbf{1}\left(z = u(x,\vec{A})\right)\right)\right),\end{aligned} \tag{65}$$

where we call $u(x,y,\vec{A},j)$ the unique configuration in Ω such that

$$u(x,y,\vec{A},j)_{V_j(\vec{A})} = y^j_{V_j(\vec{A})}\,, \qquad u(x,y,\vec{A},j)_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\,, \text{ for every } i \neq j.$$

That is, $u(x,y,\vec{A},j)$ coincides with $u(x,\vec{A})$ except that on $V_j(\vec{A})$ its content (if not empty) is taken from y^j. It follows that

$$\mathbf{1}\left(z = u(x,\vec{A})\right) = \prod_{i\,\in\,\partial\gamma} \mathbf{1}\left(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\right),$$

$$\mathbf{1}\left(z = u\left(x,y,\vec{A},j\right)\right) = \mathbf{1}\left(z_{V_j(\vec{A})} = y^j_{V_j(\vec{A})}\right) \prod_{i\,\in\,\partial\gamma:\ i\neq j} \mathbf{1}\left(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\right),$$

where $\partial\gamma$ denotes the set of leaves of γ.

As in the proof of Theorem 0.3, we let $F = F(\vec{A}, \gamma) \subset \partial\gamma$ denote the set of leaves $i \in \partial\gamma$ such that $|V_i(\vec{A})| > 1$, and recall that since μ is a product over the leaves, and p, q have the same marginals one has the identities (30) and (31). With this notation we rewrite (65) as

$$\begin{aligned}\left\langle r^{(k)}, \phi \right\rangle = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \sum_{z \in \Omega} \phi(z) \\ \times \left[\mu \left[w_{\partial\gamma} \left(Y, z, \vec{A} \right) \right] - \mu \left[w_{\partial\gamma} \left(X, z, \vec{A} \right) \right] \right. \\ \left. - \sum_{i=1}^{k} \left(\mu \left[w_{\partial\gamma} \left(X(i,y), z, \vec{A} \right) \right] - \mu \left[w_{\partial\gamma} \left(X, z, \vec{A} \right) \right] \right) \right], \quad (66)\end{aligned}$$

where $X(i,y) \in \Omega^k$ denotes the vector whose i^{th} component is Y^i while all other components are X^j. Moreover, for a given choice of $\gamma, \vec{A}, z$, the square bracket in (66) is also equal to

$$\begin{aligned}\mu \left[w_{\partial\gamma} \left(Y, z, \vec{A} \right) \right] - \mu \left[w_{\partial\gamma} \left(X, z, \vec{A} \right) \right] \\ - \sum_{i=1}^{k} \left(\mu \left[w_{\partial\gamma} \left(X(i,y), z, \vec{A} \right) \right] - \mu \left[w_{\partial\gamma} \left(X, z, \vec{A} \right) \right] \right) \\ = \mu \left[w_{F^c} \left(X, z, \vec{A} \right) \right] \ \mu \left[w_F \left(Y, z, \vec{A} \right) - w_F \left(X, z, \vec{A} \right) \right. \\ \left. - \sum_{i \in F} \left(w_F \left(X(i,y), z, \vec{A} \right) - w_F \left(X, z, \vec{A} \right) \right) \right]. \quad (67)\end{aligned}$$

In particular, if $\gamma, \vec{A}$ are such that $F = F(\gamma, \vec{A}) = \emptyset$, then (67) vanishes. Moreover, (67) vanishes also when F is a single leaf. Indeed, if e.g. $F = \{j\}$, then

$$\begin{aligned}w_F \left(Y, z, \vec{A} \right) - w_F \left(X, z, \vec{A} \right) - \sum_{i \in F} \left(w_F \left(X(i,y), z, \vec{A} \right) - w_F \left(X, z, \vec{A} \right) \right) \\ = w_F \left(Y, z, \vec{A} \right) - w_F \left(X(j,y), z, \vec{A} \right) = 0. \quad (68)\end{aligned}$$

Thus, we may restrict to the case of $|F| > 1$. Therefore,

$$\left| \left\langle r^{(k)}, \phi \right\rangle \right| \leqslant \|\phi\|_\infty \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \mathbf{1}(|F| > 1) \, \mu \left[\Gamma \left(\gamma, \vec{A}, X, Y \right) \right], \quad (69)$$

where we use the notation

$$\begin{aligned}\Gamma \left(\gamma, \vec{A}, X, Y \right) = \sum_{z \in \Omega} w_{F^c} \left(X, z, \vec{A} \right) \left| w_F \left(Y, z, \vec{A} \right) - w_F \left(X, z, \vec{A} \right) \right. \\ \left. - \sum_{i \in F} \left(w_F \left(X(i,y), z, \vec{A} \right) - w_F \left(X, z, \vec{A} \right) \right) \right|. \quad (70)\end{aligned}$$

Clearly, if $X = Y$ the expression within absolute values above vanishes. Moreover, the same applies if (X, Y) is such that $X^i = Y^i$ for all $i \in F$ except one leaf $j \in F$ where $X^j \neq Y^j$. Indeed, in this case $X(i, y) = X$ for all $i \in F$, $i \neq j$, and $X(j, y) = Y$. Given X, Y we write $E = E(X, Y) \subset F$ for the set of leaves $i \in F$ where $X^i \neq Y^i$. Then

$$\begin{aligned}
&\Gamma\left(\gamma, \vec{A}, X, Y\right) = \sum_z w_{E^c \cup F^c}\left(X, z, \vec{A}\right) \\
&\left| w_E\left(Y, z, \vec{A}\right) - w_E\left(X, z, \vec{A}\right) - \sum_{i \in E}\left(w_E\left(X(i, y), z, \vec{A}\right) - w_E\left(X, z, \vec{A}\right)\right)\right| \\
&\qquad \leqslant \sum_z w_{E^c}\left(X, z, \vec{A}\right)\left| w_E\left(Y, z, \vec{A}\right) - \sum_{i \in E} w_E\left(X(i, y), z, \vec{A}\right)\right| \\
&\qquad\qquad \mathbf{1}(|E| > 1) + (|E| - 1)\mathbf{1}(|E| > 1) \\
&\qquad\qquad \leqslant (|E| + 1)\mathbf{1}(|E| > 1) + (|E| - 1)\mathbf{1}(|E| > 1) \\
&\qquad\qquad\qquad = 2|E|\mathbf{1}(|E| > 1) \leqslant 2|F|\mathbf{1}(|E| > 1). \quad (71)
\end{aligned}$$

Next, we use $|F| \leqslant n/2$, since $\cup_{i \in F} V_i(\vec{A}) \subset [n]$ and $|V_i(\vec{A})| \geqslant 2$ for all $i \in F$. Since μ is a product over leaves and on each leaf it satisfies (28), a union bound over the set of pairs in F shows that

$$\mu\left[|E| > 1\right] \leqslant \tfrac{1}{2}\,|F|(|F| - 1)\|p - q\|_{\mathrm{TV}}^2. \quad (72)$$

Therefore,

$$\left|\left\langle r^{(k)}, \phi\right\rangle\right| \leqslant \|\phi\|_\infty \|p - q\|_{\mathrm{TV}}^2 \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\nu\left(|F|^2(|F| - 1)\right). \quad (73)$$

Next, observe that $|F| \leqslant n/2$, since $\cup_{i \in F} V_i(\vec{A}) \subset [n]$ and $|V_i(\vec{A})| \geqslant 2$ for all $i \in F$. Thus, if $|F| > 1$ one has $|F|^2(|F| - 1) \leqslant n^2(n - 2)/8$. Summarizing,

$$\left|\left\langle r^{(k)}, \phi\right\rangle\right| \leqslant \tfrac{1}{8}\, n^2(n - 2)\, \|\phi\|_\infty\, \|p - q\|_{\mathrm{TV}}^2 \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\nu(|F| > 1). \quad (74)$$

For a given tree γ, the argument in (37) shows that

$$\nu(|F| > 1) \leqslant \nu(|F| \geqslant 1) \leqslant \tfrac{1}{2}\, n(n - 1) \sum_{i=1}^k \left(\frac{r(\nu)}{2}\right)^{d_i(\gamma)} = \tfrac{1}{2}\, n(n - 1)\, \omega(\gamma),$$

where we use the fact that the event $F \geqslant 1$ coincides with $\cup_{i=1}^k \mathcal{A}_i$ from the proof of (37). This ends the proof of (64), which completes the proof of (45). □

0.7 Monotonicity of W along the nonlinear evolution

Before proving Theorem 0.2, we show that the distance W appearing in that statement is monotone along the semigroup. We refer to [42] for a similar argument in the case of kinetic models. It will be convenient to work with

a more symmetric version of (18). Since $p \circ q = \sum_A \bar{\nu}(A)(p \circ q)^A$ we may rewrite (18) as

$$C_\gamma(p) = \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \widehat{C}_\gamma^{\vec{A}}(p) \tag{75}$$

where $\nu(\vec{A})$ is defined as in (19), and $\widehat{C}_\gamma^{\vec{A}}(p)$ represents the symmetric convolutions associated to the sampled sets $(A_1, \ldots, A_{k-1})$. In other words, $\widehat{C}_\gamma^{\vec{A}}(p)$ is defined recursively by the following relations. If $\gamma \in \Gamma(k)$ then, decomposing γ into the left and right subtrees γ_l, γ_r as in (14), one has

$$\widehat{C}_\gamma^{\vec{A}}(p) = \left(\widehat{C}_{\gamma_l}^{\vec{A}_l}(p) \circ \widehat{C}_{\gamma_r}^{\vec{A}_r}(p) \right)^{A_1} \tag{76}$$

where A_1 is the set attached to the root, and

$$\vec{A}_l := (A_2, \ldots, A_j), \vec{A}_r := (A_{j+1}, \ldots, A_{k-1})$$

are the sets associated to the left and right subtrees respectively. For example, in the case where $C_\gamma(p) = ((p \circ p) \circ p) \circ p$ as in Figure 1, one has

$$C_\gamma(p) = \sum_{A_1, A_2, A_3 \subset [n]} \bar{\nu}(A_1)\bar{\nu}(A_2)\bar{\nu}(A_3) \widehat{C}_\gamma^{A_1, A_2, A_3}(p) \tag{77}$$

where $\widehat{C}_\gamma^{A_1, A_2, A_3}(p) = (((p \circ p)^{A_3} \circ p)^{A_2}) \circ p)^{A_1}$. Therefore, as in (23) one obtains the decomposition

$$p_t = \sum_{k=1}^{\infty} \beta_t(k) \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A})\, \widehat{C}_\gamma^{\vec{A}}(p). \tag{78}$$

Lemma 0.8. *For any $p, q \in \mathcal{P}(\Omega)$, any $k \in \mathbb{N}$, any $\gamma \in \Gamma(k)$, and any $\vec{A} \in \mathcal{V}_n^{k-1}$, one has*

$$W\left(\widehat{C}_\gamma^{\vec{A}}(p), \widehat{C}_\gamma^{\vec{A}}(q) \right) \leqslant W(p, q). \tag{79}$$

In particular, $W(S_t(p), S_t(q)) \leqslant W(p, q)$ for any $t \geqslant 0$.

Proof. First we show

$$W((p_1 \circ p_2), (q_1 \circ q_2)) \leqslant \tfrac{1}{2} W(p_1, q_1) + \tfrac{1}{2} W(p_2, q_2) \tag{80}$$

for all $p_1, p_2, q_1, q_2 \in \mathcal{P}(\Omega)$. For a fixed $A \subset [n]$ and configurations $x_1, x_2 \in \Omega$ we define

$$\Pi_{x_1, x_2, A}(z) := \tfrac{1}{2}\big(\mathbf{1}(z_A = x_{1,A}, z_{A^c} = x_{2,A^c}) + \mathbf{1}(z_A = x_{2,A}, z_{A^c} = x_{1,A^c}) \big), \quad z \in \Omega.$$

Now we choose $\{X_1, Y_1\}, \{X_2, Y_2\}$ random variables such that

- $\mathbb{E}[d(X_1, Y_1)] = W(p_1, q_1), \mathbb{E}[d(X_2, Y_2)] = W(p_2, q_2)$.
- X_i is distributed with p_i and Y_i is distributed with $q_i, i = 1, 2$.
- $\{X_1, Y_1\}$ and $\{X_2, Y_2\}$ are independent.

Note that

$$\mathbb{E}\left[\Pi_{X_1,X_2,A}\right] = (p_1 \circ p_2)^A. \tag{81}$$

Now, given $x_1, x_2, y_1, y_2 \in \Omega$ and $A \subset [n]$, we consider the following probability measure

$$\begin{aligned}\pi(z_1, z_2) := \tfrac{1}{2}\big(&\mathbf{1}(z_{1,A} = x_{1,A}, z_{1,A^c} = x_{2,A^c}, z_{2,A} = y_{1,A}, z_{2,A^c} = y_{2,A^c}) \\ &+ \mathbf{1}(z_{1,A} = x_{2,A}, z_{1,A^c} = x_{1,A^c}, z_{2,A} = y_{2,A}, z_{2,A^c} = y_{1,A^c})\big).\end{aligned}$$

Note that this is a coupling of $\Pi_{x_1,x_2,A}$ and $\Pi_{y_1,y_2,A}$. Therefore

$$\begin{aligned}W\left(\Pi_{x_1,x_2,A}, \Pi_{y_1,y_2,A}\right) &\leqslant \sum_{z_1,z_2} \pi(z_1, z_2) d(z_1, z_2) \\ &= \tfrac{1}{2}\big(d(x_{1,A}x_{2,A^c}, y_{1,A}y_{2,A^c}) + d(x_{2,A}x_{1,A^c}, y_{2,A}y_{1,A^c})\big) \\ &= \tfrac{1}{2}\big(d(x_1, y_1) + d(x_2, y_2)\big).\end{aligned} \tag{82}$$

Moreover, by convexity

$$W\left((p_1 \circ p_2), (q_1 \circ q_2)\right) \leqslant \sum_{A \subset [n]} \nu(A) W\left((p_1 \circ p_2)^A, (q_1 \circ q_2)^A\right) \tag{83}$$

and, for each $A \subset [n]$,

$$\begin{aligned}&W\left((p_1 \circ p_2)^A, (q_1 \circ q_2)^A\right) \\ &\qquad \leqslant \mathbb{E}\left[W\left(\Pi_{X_1,X_2,A}, \Pi_{Y_1,Y_2,A}\right)\right] = \tfrac{1}{2} W(p_1, q_1) + \tfrac{1}{2} W(p_2, q_2).\end{aligned} \tag{84}$$

Combining the last two inequalities we obtain (80).

We now prove (79) by induction over $k \geqslant 1$. Clearly, $k = 1$ is trivial. The case $k = 2$ follows by (84). Suppose that (79) holds for any $j \leqslant k-1$, $\gamma \in \Gamma(j)$, $\vec{A} \in \mathcal{V}_n^{j-1}$ and let A_1 be the set attached to the root of γ. If $\gamma \in \Gamma(k)$ then we decompose γ into the left and right subtrees γ_l, γ_r as in (14), so that

$$\begin{aligned}\widehat{C}_\gamma^{\vec{A}}(p) &= \left(\widehat{C}_{\gamma_l}^{\vec{A}_l}(p) \circ \widehat{C}_{\gamma_r}^{\vec{A}_r}(p)\right)^{A_1} \\ \widehat{C}_\gamma^{\vec{A}}(q) &= \left(\widehat{C}_{\gamma_l}^{\vec{A}_l}(q) \circ \widehat{C}_{\gamma_r}^{\vec{A}_r}(q)\right)^{A_1}.\end{aligned}$$

where $\vec{A}_l := (A_2, \ldots, A_j)$ and $\vec{A}_r := (A_{j+1}, \ldots, A_{k-1})$, and j is the number of leaves of γ_l. Then, using (84) again and the inductive hypothesis we have

$$\begin{aligned}W\left(\widehat{C}_\gamma^{\vec{A}}(p), \widehat{C}_\gamma^{\vec{A}}(q)\right) &= W\left(\left(\widehat{C}_{\gamma_l}^{\vec{A}_l}(p) \circ \widehat{C}_{\gamma_r}^{\vec{A}_r}(p)\right)^{A_1}, \left(\widehat{C}_{\gamma_l}^{\vec{A}_l}(q) \circ \widehat{C}_{\gamma_r}^{\vec{A}_r}(q)\right)^{A_1}\right) \\ &\leqslant \tfrac{1}{2} W\left(\widehat{C}_{\gamma_l}^{\vec{A}_l}(p), \widehat{C}_{\gamma_l}^{\vec{A}_l}(q)\right) + \tfrac{1}{2} W\left(\widehat{C}_{\gamma_r}^{\vec{A}_r}(p), \widehat{C}_{\gamma_r}^{\vec{A}_r}(q)\right) \leqslant W(p, q).\end{aligned}$$

This proves (79).

By (75) and convexity,

$$W(C_\gamma(p), C_\gamma(q)) \leqslant W(p, q). \tag{85}$$

Finally, using the expression (78) for $S_t(p) = p_t$, again by convexity we obtain

$$W(S_t(p), S_t(q)) \leqslant W(p, q),$$

for all $t \geqslant 0$. □

Remark 0.9. *We note that* (80) *shows in particular that*

$$W\left(p\circ p, q\circ q\right) \leqslant W\left(p,q\right). \tag{86}$$

This monotonicity is not satisfied by the total variation distance. For example let us consider $\Omega=\{0,1\}^2$, $p:=\mathbf{1}_{(1,1)}$, $q:=\frac{1}{2}(\mathbf{1}_{(1,1)}+\mathbf{1}_{(0,0)})$ *and suppose* ν *is the uniform crossover. Then one can check that* $p\circ p=\mathbf{1}_{(1,1)}$ *and* $q\circ q=\frac{3}{8}(\mathbf{1}_{(1,1)}+\mathbf{1}_{(0,0)})+\frac{1}{8}(\mathbf{1}_{(1,0)}+\mathbf{1}_{(0,1)})$, *and therefore* $\|p\circ p-q\circ q\|_{\mathrm{TV}}=\frac{5}{8}>\frac{1}{2}=\|p-q\|_{\mathrm{TV}}$.

0.8 Proof of Theorem 0.2

For $k\in\mathbb{N}$, the moment measures $F_t^{k,N}\in\mathcal{P}(\Omega^k)$ and $\bar{F}_t^{k,N}\in\mathcal{P}(\Omega^k)$ are defined by

$$F_t^{k,N}=\sum_{\eta\,\in\,\Omega^N}\mu_{N,t}(\eta)(\lambda_\eta)^{\otimes k},\qquad \bar{F}_t^{k,N}=\sum_{\eta\,\in\,\Omega^N}\mu_N(\eta)(S_t\lambda_\eta)^{\otimes k}. \tag{87}$$

By the triangular inequality we have

$$\left|P_k\mu_{N,t}(\phi_k)-p_t^{\otimes k}(\phi_k)\right| \leqslant \mathcal{T}_1+\mathcal{T}_2+\mathcal{T}_3\,, \tag{88}$$

where

$$\begin{gathered}\mathcal{T}_1:=\left|P_k\mu_{N,t}(\phi_k)-F_t^{k,N}(\phi_k)\right|\,,\quad \mathcal{T}_2:=\left|F_t^{k,N}(\phi_k)-\bar{F}_t^{k,N}(\phi_k)\right|\,,\\ \mathcal{T}_3:=\left|\bar{F}_t^{k,N}(\phi_k)-p_t^{\otimes k}(\phi_k)\right|.\end{gathered} \tag{89}$$

We are going to estimate each term separately.

The term $\mathcal{T}_1$ can be estimated by using a simple combinatorial argument, and one obtains

$$\mathcal{T}_1\leqslant\frac{2k(k-1)}{N}\|\phi_k\|_\infty. \tag{90}$$

Since the proof is identical to that in [9, Lemma 3.11] we omit the details.

The last term $\mathcal{T}_3$ is estimated using the initial chaos. Note that

$$p_t^{\otimes k}\left(\phi_k\right)-\bar{F}_t^{k,N}\left(\phi_k\right)=\sum_{\eta\,\in\,\Omega^N}\mu_N(\eta)\left(\prod_{i=1}^k\left\langle S_t(p),\phi^i\right\rangle-\prod_{i=1}^k\left\langle S_t(\lambda_\eta),\phi^i\right\rangle\right). \tag{91}$$

Using $\prod_{i=1}^k a_i-\prod_{i=1}^k b_i=\sum_{i=1}^k(\prod_{1\,\leqslant\,j<i}a_j)(a_i-b_i)(\prod_{i<j\,\leqslant\,k}b_j)$, we obtain

$$\begin{aligned}\mathcal{T}_3 &\leqslant \sum_{\eta\,\in\,\Omega^N}\mu_N(\eta)\sum_{i=1}^k\left(\prod_{1\,\leqslant\,j<i}\left|\left\langle S_t(p),\phi^j\right\rangle\right|\right)\\ &\qquad\left|\left\langle S_t(p)-S_t(\lambda_\eta),\phi^i\right\rangle\right|\left(\prod_{i<j\,\leqslant\,k}\left|\left\langle S_t(\lambda_\eta),\phi^j\right\rangle\right|\right)\\ &\leqslant 2k\|\phi_k\|_\infty\,\mu_N\left[W\left(S_t(p),S_t(\lambda_{\chi^N})\right)\right]\leqslant 2k\|\phi_k\|_\infty\,\mu_N\left[W\left(p,\lambda_{\chi^N}\right)\right],\end{aligned}$$

where χ^N has distribution μ_N, and we have used

$$|\langle \mu - \mu', f\rangle| \leqslant 2\|f\|_\infty \|\mu - \mu'\|_{\mathrm{TV}} \leqslant 2\|f\|_\infty W(\mu,\mu'), \tag{92}$$

for any $\mu, \mu' \in \mathcal{P}(\Omega)$, $f : \Omega \mapsto \mathbb{R}$, and the monotonicity from Lemma 0.8.

The estimate of the second term $\mathcal{T}_2$ is more delicate. Here we use the contraction proved in Theorem 0.3 and Theorem 0.5. Define, for $s \in [0,t]$,

$$\Psi_s^k = \sum_{\eta \in \Omega^N} \mu_{N,t-s}(\eta)(S_s(\lambda_\eta))^{\otimes k}.$$

Then $\Psi_s^k \in \mathcal{P}(\Omega^k)$ for all $s \in [0,t]$, and

$$F_t^{k,N}(\phi_k) = \Psi_0^k(\phi_k)\,, \qquad \bar{F}_t^{k,N}(\phi_k) = \Psi_t^k(\phi_k).$$

Therefore,

$$\bar{F}_t^{k,N}(\phi_k) - F_t^{k,N}(\phi_k) = \int_0^t \partial_s \left[\Psi_s^k(\phi_k)\right]\, ds\,. \tag{93}$$

Now,

$$\begin{aligned}
&\partial_s \left[\Psi_s^k(\phi_k)\right] \\
&\quad = \sum_{\eta \in \Omega^N} (\partial_s \mu_{N,t-s}(\eta))(S_s(\lambda_\eta))^{\otimes k}(\phi_k) + \sum_{\eta \in \Omega^N} \mu_{N,t-s}(\eta)\partial_s (S_s(\lambda_\eta))^{\otimes k}(\phi_k) \\
&\qquad\qquad = -\sum_{\eta \in \Omega^N} \mu_{N,t-s}(\eta)\mathcal{L}_N \prod_{\ell=1}^k \left\langle S_s(\lambda_\eta), \phi^\ell\right\rangle \\
&\qquad\qquad + \sum_{\eta \in \Omega^N} \mu_{N,t-s}(\eta) \sum_{i=1}^k \left\langle Q(S_s(\lambda_\eta)), \phi^i\right\rangle \prod_{\ell \neq i} \left\langle S_s(\lambda_\eta, \phi^\ell\right\rangle\,.
\end{aligned} \tag{94}$$

Next, we show that

$$Q(S_s(\lambda_\eta)) = \frac{1}{N}\sum_{i<j}\sum_A \nu(A)\bar{S}_s(\lambda_\eta)\left(\lambda_{\eta^{i,j,A}} - \lambda_\eta\right). \tag{95}$$

Consider the linearized equation (43). Taking $q \in \mathcal{P}(\Omega)$ and $h = Q(q)$ one has that the solution $h_t = \bar{S}_t(q)(h)$ satisfies $h_t = Q\left(S_t(q)\right)$, that is

$$\bar{S}_t(q)(Q(q)) = Q\left(S_t(q)\right)\,, \qquad t \geqslant 0, \tag{96}$$

for all $q \in \mathcal{P}(\Omega)$. To see this note that for all $t \geqslant 0$,

$$\sum_{\sigma \in \Omega} S_t(q)(\sigma) = 1\,, \qquad \sum_{\sigma \in \Omega} Q(S_t(q))(\sigma) = 0\,,$$

and therefore

$$
\begin{aligned}
&\partial_t Q\left(S_t(q)\right)(\sigma') \\
&= \sum_{\substack{A\subset[n]\\ \sigma\in\Omega}} \nu(A)\Big(Q(S_t(q))(\sigma_{A^c}\sigma'_A)S_t(q)(\sigma_A\sigma'_{A^c}) \\
&\quad + S_t(q)(\sigma_{A^c}\sigma'_A)Q(S_t(q))(\sigma_A\sigma'_{A^c}) - Q(S_t(q))(\sigma')S_t(q)(\sigma)\Big) \\
&= \sum_{\substack{A\subset[n]\\ \sigma\in\Omega}} \nu(A)\Big(Q(S_t(q))(\sigma_{A^c}\sigma'_A)S_t(q)(\sigma_A\sigma'_{A^c}) + S_t(q))(\sigma_{A^c}\sigma'_A)Q(S_t(q))(\sigma_A\sigma'_{A^c}) \\
&\quad - Q(S_t(q))(\sigma')S_t(q)(\sigma) - Q(S_t(q))(\sigma)S_t(q)(\sigma')\Big)2\widehat{Q}(S_t(q),Q(S_t(q)))(\sigma').
\end{aligned}
$$

This proves (96). If $q=\lambda_\eta$ for some $\eta\in\Omega^N$, then

$$
Q(\lambda_\eta) = \sum_A \nu(A)\left((\lambda_\eta)_A\otimes(\lambda_\eta)_{A^c} - \lambda_\eta\right). \tag{97}
$$

Moreover, for any $A\subset[n]$,

$$
\begin{aligned}
(\lambda_\eta)_A\otimes(\lambda_\eta)_{A^c} - \lambda_\eta &= \frac{1}{2N^2}\sum_{i,j=1}^N \left(\mathbf{1}_{\eta^{i,j,A}(i)} + \mathbf{1}_{\eta^{i,j,A}(j)} - \mathbf{1}_{\eta(i)} - \mathbf{1}_{\eta(j)}\right) \\
&= \frac{1}{N}\sum_{i<j}\left(\lambda_{\eta^{i,j,A}} - \lambda_\eta\right).
\end{aligned} \tag{98}
$$

It follows that

$$
Q(\lambda_\eta) = \frac{1}{N}\sum_{i<j}\sum_A \nu(A)(\lambda_{\eta^{i,j,A}} - \lambda_\eta). \tag{99}
$$

Thus, using (96) and the linearity of $\bar{S}_s(\lambda_\eta)(\cdot)$ we obtain (95).

On the other hand,

$$
\begin{aligned}
&\mathcal{L}_N\prod_{\ell=1}^k\left\langle S_s(\lambda_\eta),\phi^\ell\right\rangle \\
&\qquad = \frac{1}{N}\sum_{i<j}\sum_A \nu(A)\left(\prod_{\ell=1}^k\left\langle S_s(\lambda_{\eta^{i,j,A}}),\phi^\ell\right\rangle - \prod_{\ell=1}^k\left\langle S_s(\lambda_\eta),\phi^\ell\right\rangle\right).
\end{aligned} \tag{100}
$$

Suppose we can show that

$$
\begin{aligned}
&\Bigg|\prod_{\ell=1}^k\left\langle S_s(\lambda_{\eta^{i,j,A}}),\phi^\ell\right\rangle - \prod_{\ell=1}^k\left\langle S_s(\lambda_\eta),\phi^\ell\right\rangle \\
&\qquad - \sum_{u=1}^k\left\langle\bar{S}_s(\lambda_\eta)(\lambda_{\eta^{i,j,A}}-\lambda_\eta),\phi^u\right\rangle\prod_{\ell\neq u}\left\langle S_s(\lambda_\eta,\phi^\ell\right\rangle\Bigg| \leqslant K_s,
\end{aligned} \tag{101}
$$

for some function K_s, $s \in [0,\infty)$, independent of η, i, j, A. Then by (94) and (95) we would have $\mathcal{T}_2 \leqslant \frac{N}{2}\int_0^t K_s ds$. Thus the proof will be completed by proving (101) for a suitable K_s.

Observe that

$$\begin{aligned}
&\prod_{\ell=1}^{k} a_\ell - \prod_{\ell=1}^{k} b_\ell - \sum_{u=1}^{k} c_u \prod_{j\neq u} b_j \\
&= \sum_{u=1}^{k}\left[\left(\prod_{1\leqslant j<u} a_j\right)(a_u-b_u) - \left(\prod_{1\leqslant j<u} b_j\right)c_u\right]\prod_{j>u} b_j \\
&= \sum_{u=1}^{k}(a_u-b_u-c_u)\prod_{j\neq u} b_j + \sum_{u=1}^{k}\left[\left(\prod_{1\leqslant j<u} a_j\right) - \left(\prod_{1\leqslant j<u} b_j\right)\right](a_u-b_u)\prod_{j>u} b_j \\
&= \sum_{u=1}^{k}(a_u-b_u-c_u)\prod_{j\neq u} b_j + \sum_{u=1}^{k}\sum_{v=1}^{u-1}\left(\prod_{j<v} a_j\right)\left(\prod_{v<j<u} b_j\right)(a_v-b_v)(a_u-b_u)\prod_{j>u} b_j.
\end{aligned} \tag{102}$$

Define $a_\ell = \langle S_s(\lambda_{\eta^{i,j,A}}), \phi^\ell\rangle$, $b_\ell = \langle S_s(\lambda_\eta), \phi^\ell\rangle$, and $c_\ell = \langle \bar{S}_s(\lambda_\eta)(\lambda_{\eta^{i,j,A}} - \lambda_\eta), \phi^\ell\rangle$. Noticing that

$$\|\lambda_{\eta^{i,j,A}} - \lambda_\eta\|_{\mathrm{TV}} \leqslant \frac{2}{N}, \tag{103}$$

and that $\lambda_{\eta^{i,j,A}}, \lambda_\eta$ have the same marginals, it follows from Theorem 0.5 that

$$|a_u - b_u - c_u| \leqslant n^5 \|\phi^u\|_\infty e^{-D(\nu)\, s}\frac{4}{N^2}. \tag{104}$$

Moreover, Theorem 0.3 shows that for all u, v,

$$|a_v - b_v||a_u - b_u| \leqslant n^4 \|\phi^v\|_\infty \|\phi^u\|_\infty e^{-2D(\nu)\, s}\frac{4}{N^2}. \tag{105}$$

These bounds hold uniformly in $\eta \in \Omega^N, 1 \leqslant i < j \leqslant n, A \subset [n]$. This proves (101) with

$$K_s = k e^{-D(\nu)\, s}\left(n^5 + (k-1)n^4 e^{-D(\nu)\, s}\right)\frac{\|\phi_k\|_\infty}{N^2}. \tag{106}$$

Therefore,

$$\mathcal{T}_2 \leqslant \frac{N}{2}\int_0^t K_s \leqslant \frac{k^2 n^5}{D(\nu)N}\|\phi_k\|_\infty\left(1 - e^{-D(\nu)\, t}\right). \tag{107}$$

The proof of Theorem 0.2 is complete.

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